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F. Count the Numbers

time limit per test: 1 second
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given a sorted array a of n elements, and some queries. In each query, you are given a pair $[x, y]$ and you have to count how many numbers a_i are there such that $x \leq a_i \leq y$. For example, if the array is $[10, 20, 20, 45, 79]$ and you are given a query $[20, 50]$, then answer will be 3 because there are in total 3 numbers that's value is between 20 and 50.

Input

The first line of the input contains $n (1 \leq n \leq 10^5)$ and $q (1 \leq q \leq 10^5)$ denoting the array size and the number of queries respectively. The next line will contain the array elements separated by space where $1 \leq a_i \leq 10^9$ where $i = 0, 1, 2, \ldots, n-1$. Each of the next q lines will contain a pair $[x, y]$ where $1 \leq x \leq y \leq 10^9$. See the sample input format for better understanding.

Note1: It is guaranteed that the given array is sorted in **non-decreasing** order.

Note2: It is also guaranteed that the queries are valid. Which means, for each query $[x, y]$, $x \leq y$.

Output

For each query $[x, y]$, output a single integer P denoting the number of elements in the array a such that $x \leq a_i \leq y$.

Example

input Copy

```
5 3
10 20 20 45 79
20 50
5 45
1 100
```

output Copy

```
3
4
5
```

CSE221 Assignment 02 Fall 2026
Contest is running
3 days
Contestant

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Language: Java 21 64bit

Choose file: Choose File No file chosen

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